

a)

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~ | G phanthawas-dev < k3d-sds-cluster < 22:46:36
) kubectl describe deployment nginx-deployment
Name:          nginx-deployment
Namespace:     default
CreationTimestamp: Wed, 10 Sep 2025 22:41:19 +0700
Labels:        <none>
Annotations:   deployment.kubernetes.io/revision: 1
Selector:      app=nginx
Replicas:      2 desired | 2 updated | 2 total | 2 available | 0 unavailable
StrategyType:  RollingUpdate
MinReadySeconds: 0
RollingUpdateStrategy: 25% max unavailable, 25% max surge
Pod Template:
  Labels:  app=nginx
  Containers:
    nginx:
      Image:   nginx:1.14.2
      Port:    80/TCP
      Host Port: 0/TCP
      Environment: <none>
      Mounts:     <none>
      Volumes:     <none>
      Node-Selectors: <none>
      Tolerations:  <none>
Conditions:
  Type           Status  Reason
  ----           -
  Available      True    MinimumReplicasAvailable
  Progressing    True    NewReplicaSetAvailable
OldReplicaSets: <none>
NewReplicaSet:  nginx-deployment-d556bf558 (2/2 replicas created)
Events:
  Type           Reason             Age   From                  Message
  ----           -
  Normal         ScalingReplicaSet   5m18s deployment-controller Scaled up replica set nginx-deployment-d556bf558 to 2
~ | G phanthawas-dev < k3d-sds-cluster < 22:46:37
)
```

b)

deployment.yaml

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: todo-redis-deployment
  labels:
    app: todo-redis
spec:
  replicas: 1
  selector:
    matchLabels:
      app: todo-redis
  template:
    metadata:
      labels:
        app: todo-redis
```

```

spec:
  containers:
  - name: todo
    image: natawut/todo-service:release-2.1
    ports:
    - containerPort: 8000
    env:
    - name: REDIS_HOST
      value: "localhost"
  - name: redis
    image: redis:alpine
    ports:
    - containerPort: 6379
---
apiVersion: v1
kind: Service
metadata:
  name: todo-service
spec:
  selector:
    app: todo-redis
  ports:
  - protocol: TCP
    port: 80
    targetPort: 8000
  type: ClusterIP

```

ingress.yaml

```

apiVersion: networking.k8s.io/v1
kind: Ingress
metadata:
  name: todo-ingress
  annotations:
    nginx.ingress.kubernetes.io/rewrite-target: /
spec:
  rules:
  - host: localhost
    http:
      paths:
      - path: /
        pathType: Prefix
        backend:
          service:
            name: todo-service
            port:
              number: 80

```

c)

```
~
G phanthawas-dev < k3d-sds-cluster < 22:49:46
} kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
nginx-deployment-d556bf558-h8fz8    1/1     Running   0           8m36s
nginx-deployment-d556bf558-m8qhg    1/1     Running   0           8m36s
todo-redis-deployment-688b4dc9f5-fcmgk 2/2     Running   0           6m4s

~
G phanthawas-dev < k3d-sds-cluster < 22:49:55
} curl -X POST -d '{"title":"todo-1","detail":"the first todo","duedate":"2024-10-13 11:00:00","tags":[],"completed":false}' http://localhost
{"id":1}

~
G phanthawas-dev < k3d-sds-cluster < 22:50:29
} curl localhost
[{"id":0,"title":"todo-1","detail":"the first todo","completed":false,"duedate":"2024-10-13T11:00:00","tags":[]}, {"id":1,"title":"todo-1","detail":"the first todo","completed":false,"duedate":"2024-10-13T11:00:00","tags":[]}]

~
G phanthawas-dev < k3d-sds-cluster < 22:50:37
}

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G phanthawas-dev < k3d-sds-cluster < 22:50:45
}

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G phanthawas-dev < k3d-sds-cluster < 22:50:46
}

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G phanthawas-dev < k3d-sds-cluster < 22:50:46
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G phanthawas-dev < k3d-sds-cluster < 22:50:46
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G phanthawas-dev < k3d-sds-cluster < 22:50:46
}

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G phanthawas-dev < k3d-sds-cluster < 22:50:47
}
```

```
~ | G phanthawas-dev < k3d-sds-cluster < 22:52:27
) kubectl get deployments
  kubectl get services
  kubectl get pods
NAME                READY   UP-TO-DATE   AVAILABLE   AGE
nginx-deployment    2/2     2             2           11m
todo-redis-deployment 1/1     1             1           10m
NAME                TYPE        CLUSTER-IP   EXTERNAL-IP   PORT(S)    AGE
kubernetes          ClusterIP   10.43.0.1     <none>        443/TCP    11m
todo-service        ClusterIP   10.43.11.17   <none>        80/TCP     10m
NAME                READY   STATUS    RESTARTS   AGE
nginx-deployment-d556bf558-h8fz8 1/1     Running   0           11m
nginx-deployment-d556bf558-m8qhg 1/1     Running   0           11m
todo-redis-deployment-688b4dc9f5-fcmgk 2/2     Running   0           8m37s

~ | G phanthawas-dev < k3d-sds-cluster < 22:52:28
) kubectl get ingress
  kubectl describe ingress todo-ingress
NAME                CLASS    HOSTS      ADDRESS    PORTS    AGE
todo-ingress        traefik  localhost  172.18.0.3  80       9m26s
Name:                todo-ingress
Labels:               <none>
Namespace:            default
Address:              172.18.0.3
Ingress Class:        traefik
Default backend:      <default>
Rules:
  Host      Path  Backends
  ----      -
  localhost /     todo-service:80 (10.42.0.12:8000)
Annotations:  nginx.ingress.kubernetes.io/rewrite-target: /
Events:       <none>

~ | G phanthawas-dev < k3d-sds-cluster < 22:52:35
)
```